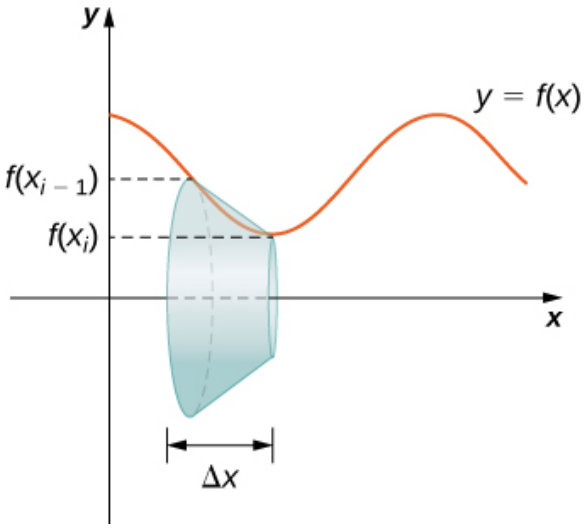
8.2 Area of a Surface of Revolution

Surface Area

Rotating a curve about a line, we obtain a surface of revolution. To find the area of this surface, we slice it into frustum bands, each has the area $2\pi\frac{r_1+r_2}{2}L$, where r_1, r_2 are the lower and upper radii and L is the slant height.



Compute the area of one band:

$$2\pi \underbrace{f(x_i^{**})}_{\text{by IVT}} \underbrace{\sqrt{1+[f'(x_i^*)]^2}\,\Delta x}_{L}$$

Add up the bands and take the limit to find the surface area:

Rotate around the x-axis.

$$S = \lim_{n\to\infty}\sum_{i=1}^{n} 2\pi f(x_i^{**})\sqrt{1+[f'(x_i^*)]^2}\,\Delta x = \int_a^b 2\pi f(x)\sqrt{1+[f'(x)]^2}\,dx$$

Rotate around the y-axis.

$$S = \int_c^d 2\pi g(y)\sqrt{1+[g'(y)]^2}\,dy$$

In general:

$$S = \int 2\pi r\,ds$$

Problem 1. Let $f(x) = \sqrt{x}$ on $[1, 4]$. Find the surface area of the surface generated by revolving the graph of $f(x)$ about the x-axis.

Solution.

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$\begin{aligned} S &= \int_1^4 2\pi\sqrt{x}\,\sqrt{1+\frac{1}{4x}}\,dx \\ &= \int_1^4 2\pi\sqrt{x+\frac{1}{4}}\,dx \\ &= 2\pi\cdot\frac{2}{3}\left(x+\frac{1}{4}\right)^{\frac{3}{2}}\Bigg|_1^4 \\ &= \frac{\pi}{6}\left(17\sqrt{17}-5\sqrt{5}\right) \end{aligned}$$

Problem 2. Consider $y = \sqrt[3]{3x}$ for $0 \le y \le 2$. Find the surface area of the surface generated by revolving the graph of $f(x)$ about the y-axis.

Solution.

$$y^3 = 3x, \qquad x = \frac{y^3}{3}, \qquad \frac{dx}{dy} = y^2$$

$$\begin{aligned} S &= \int_0^2 2\pi\,\frac{y^3}{3}\sqrt{1+y^4}\,dy \\ &= \int_0^2 \frac{\pi}{6}\,4y^3\sqrt{1+y^4}\,dy \\ &= \frac{\pi}{6}\cdot\frac{2}{3}\left(1+y^4\right)^{\frac{3}{2}}\Bigg|_0^2 \\ &= \frac{\pi}{9}\left(17^{\frac{3}{2}}-1\right) \end{aligned}$$

Problem 3. The curve $y = \sqrt{4-x^2}$, $-1 \le x \le 1$, is an arc of the circle $x^2+y^2 = 4$. Find the area of the surface obtained by rotating this arc about the x-axis.

Solution.

$$y' = \frac{-2x}{2\sqrt{4-x^2}} = \frac{-x}{\sqrt{4-x^2}}$$

$$\begin{aligned} S &= \int_{-1}^1 2\pi\sqrt{4-x^2}\,\sqrt{1+\frac{x^2}{4-x^2}}\,dx \\ &= \int_{-1}^1 2\pi\sqrt{4-x^2+x^2}\,dx \\ &= \int_{-1}^1 4\pi\,dx \\ &= 8\pi \end{aligned}$$

Review

Area between curves:

$$\int_a^b f(x) - g(x)\,dx \qquad \text{top} - \text{bottom}$$

$$\int_c^d f(y) - g(y)\,dy \qquad \text{right} - \text{left}$$

Volume:

Disk/washer:	$\int_a^b \pi\left(R^2 - r^2\right)dx$	rotation around the x-axis
Shell:	$\int_a^b 2\pi x f(x)\,dx$	rotation around the y-axis

Average value:

$$f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx$$